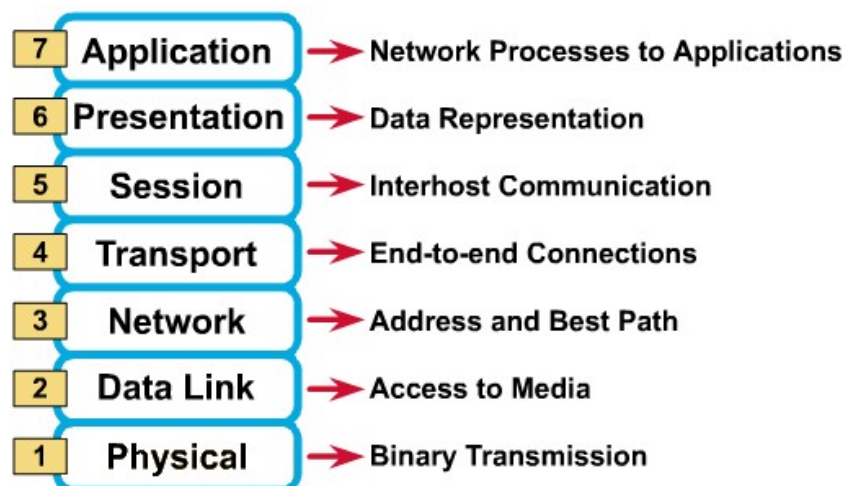


Lower Layers

1

1

L1 – L2



2

2

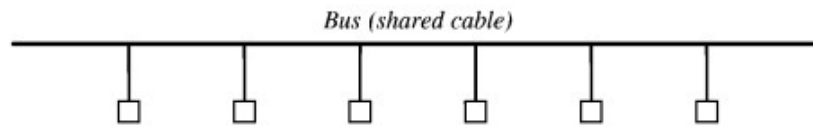
Physical Layer sublayers

- **PCS: Physical Coding Sublayer**
 - Block encoding
- **PMA: Physical Medium Attachment sublayer**
 - framing, octet synchronization/detection, and scrambling/descrambling
- **PMD: Physical Medium Dependent sublayer**
 - Adaption to the physical medium

Network Topology

- Specifies general “shape” of a network
- Handful of broad categories
- Often applied to LAN
- Primarily refers to interconnections
- Hides details of actual devices

Bus Topology



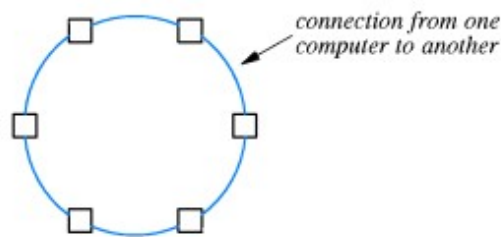
- Shared medium forms main interconnect
- Each node has a connection to the medium

Bus Topology

- Advantages
 - Works well for small networks.
 - Relatively inexpensive to implement.
 - Easy to add and remove nodes.
 - Failure of one station does not affect others.
- Disadvantages
 - Management costs can be high.
 - Potential for congestion with network traffic.
 - Performance degrades as additional nodes are added.
 - Performance degrades as cable length increases.

Ring Topology

- No central facility
- Connections go directly from one node to another



Advantages of Ring Topology

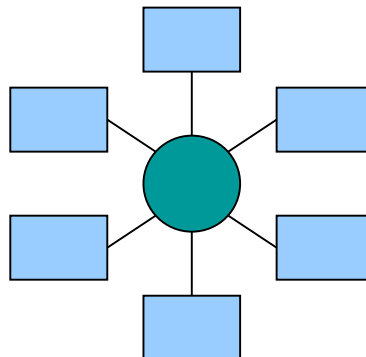
- Easier to manage; easier to locate a defective node or cable problem.
- Well-suited for transmitting signals over long distances on a LAN.
- Well-suited for Real-time / Synchronous / Time Triggered communication.
- Handles high-volume network traffic.
- Enables reliable communication.
- Additional nodes do not affect the performance of network.

Disadvantages of Ring Topology

- Expensive
- Requires more cable and network equipment
- Each node must be active.
 - A failure in a single node turns all network down.
- Each packet of data must pass through all nodes
 - Unavoidable delay.
- Not used as widely as bus topology
 - Fewer equipment options
 - Fewer options for expansion to high-speed communication

Star Topology

- Central component of network known as hub
- Each node has a separate connection to a central repeater.



Star

- Any single cable connects only two devices
 - A cabling problem affects a single node
 - More fault-tolerant
- Requires more cabling than ring or bus networks
 - Expensive
- Easily moved, isolated, or interconnected with other networks
 - Scalable
- Central point is a single point of failure.

Data Link sublayers

- The data link layer may be divided into two sublayers:
 - The LLC (Logical Link Control) :
 - multiplexes protocols running atop the data link layer,
 - Optionally: provides flow control,
 - acknowledgment, and error notification.
 - The MAC (Media Access Control) :
 - Controls access to the shared transmission medium.
 - Optionally: provides addressing.

MAC Addressing

- Identification of the parts involved in a transaction
- Direct addressing
 - The sender and receiver(s) are explicitly identified in every transaction, using physical addresses (as in Ethernet)
- Indirect (source) addressing
 - The message contents are explicitly identified (e.g. temperature of sensor X). Receivers that need the message, retrieve it from the network (as in CAN and WorldFIP)
- Indirect (time-based) addressing
 - The message is identified by the time instant at which it is transmitted (as in TTP)

MAC sublayer

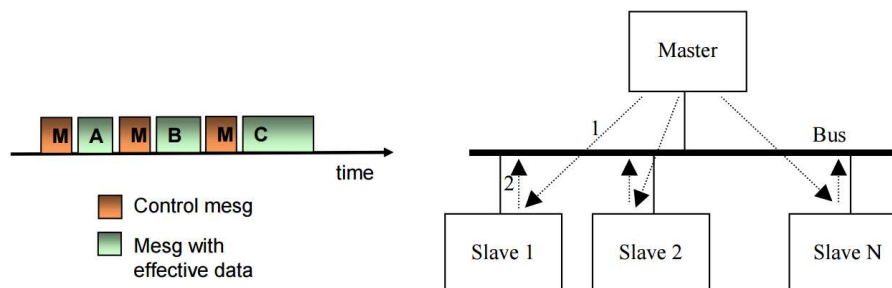
- Determines the order of network access by contending nodes and thus the network access delay
- Very important for the real-time behavior of networks that use a shared medium.

Multiple Access Protocol

- Single shared broadcast channel
 - Avoid having multiple nodes speaking at once
 - Otherwise, collisions lead to garbled data
- Multiple access protocol
 - Distributed algorithm for sharing the channel
 - Algorithm determines which node can transmit
- Classes of techniques
 - Channel partitioning: divide channel into pieces
 - Taking turns: passing a token for the right to transmit
 - Random access: allow collisions, and then recover

Medium access control - Master-slave

- Master-slave
 - Access granted by the Master
 - Nodes synchronized with the master
 - Requires one control message per data message

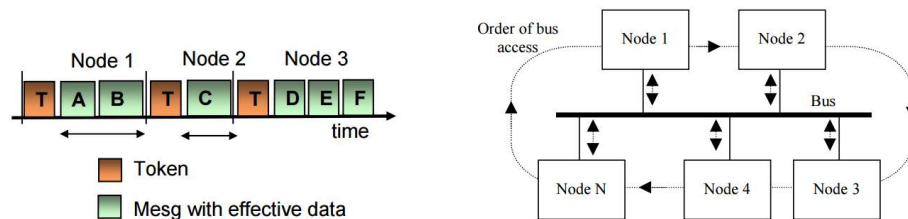


Medium access control - Master-slave

- The traffic scheduling problem becomes local to the master
- Good flexibility
- For high reliability the master must be replicated
- Master messages are natural synchronization points
- Supports precise TX triggering
 - Att: reception instants may vary from node to node, depending on the distance to the master
- Examples: WorldFIP, Ethernet Powerlink, Bluetooth within each piconet

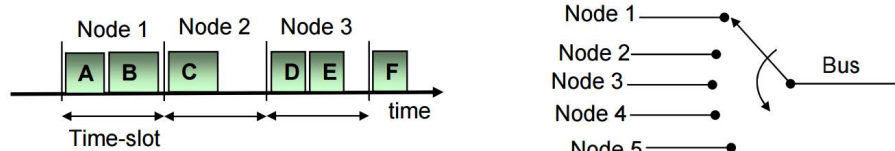
Medium access control – Token-passing

- Medium access control – Token-passing
- Access granted by the possession of the token
- Order of access enforced by token circulation
- Asynchronous bus access (generally impossible to determine a priori the exact access instants)
- Real-time operation requires bounded token holding time
- Ex: IEEE 802.4, FDDI, PROFIBUS



MAC - TDMA

- Time-Division Multiple Access
- Access granted in dedicated time-slot
- Time-slots are pre-defined in a cyclic framework
- Tight synchronization with bus time (bus access instants are predetermined)
- Requires global (clock) synchronization



MAC - TDMA

- Addressing can be based on time
- High data efficiency
- Quality of synchronization bounds efficiency (length of guarding windows between slots)
- Typically uses static table-based scheduling
- Ex: TTP/C, FlexRay-sync, TT-CAN, PROFINET

Random Access Protocols

- When a node has packet to send
 - Transmit at full channel data rate R .
 - No a priori coordination among nodes
- Two or more transmitting nodes → “collision”
- Random access MAC protocol specifies:
 - How to detect collisions
 - How to recover from collisions
- Examples
 - CSMA/CD (Ethernet: IEEE 802.3)
 - CSMA/CA (Wireless: IEEE 802.11)

Key Ideas of Random Access

- Carrier sense
 - *Listen before speaking, and don't interrupt*
 - Checking if someone else is already sending data
 - ... and waiting till the other node is done
- Collision detection
 - *If someone else starts talking at the same time, stop*
 - Realizing when two nodes are transmitting at once
 - ...by detecting that the data on the wire is garbled
- Randomness
 - *Don't start talking again right away*
 - Waiting for a random time before trying again

MAC - CSMA

- Carrier-Sense Multiple Access
- Set of protocols based on sensing bus inactivity before transmitting (asynchronous bus access)
- There may be collisions
- Upon collision, nodes back off and retry later, according to some specific rule (this rule determines, to a large extent, the real-time features of the protocol)

MAC – CSMA/CD

- CSMA/CD Carrier-Sense Multiple Access with Collision Detection
- Used in shared Ethernet
- Collisions are destructive and are detected within collision windows (slots)
- Upon collision, the retry interval is random and the randomization window is doubled for each retry until 1024 slots (Binary Exponential Back-off)
- Non-deterministic (particularly with chained collisions)

MAC: CSMA/DCR

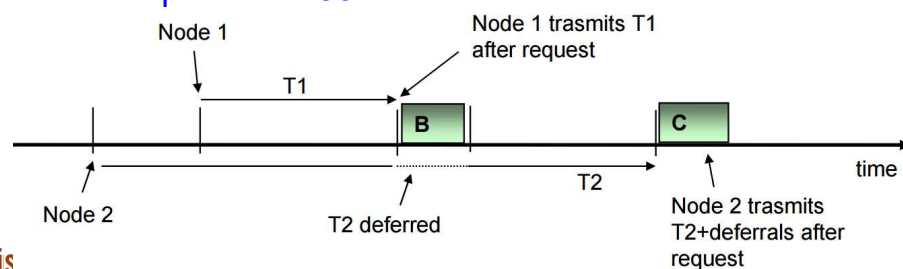
- CSMA/DCR Carrier-Sense Multiple Access with Deterministic Collision Resolution
- Collisions are destructive
- Back off and retry based on priority (binary tree search)
- Deterministic
- Ex. G. Le Lann's modified Ethernet

MAC - CSMA/BA

- CSMA/BA Carrier-Sense Multiple Access with Bit-wise Arbitration
- Bit-wise arbitration on collisions.
- Collision is non destructive
- Upon collision, highest priority node is unaffected. Nodes with lower priorities retry right after.
- Deterministic
- Exemple: CAN

MAC – CSMA/CA Async

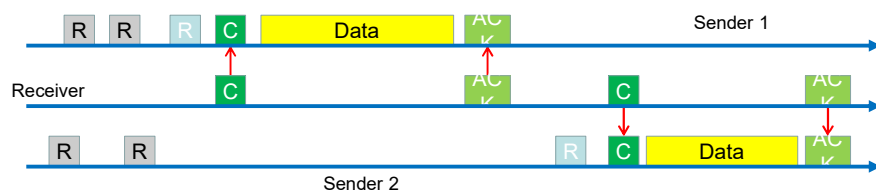
- CSMA/CA: Carrier-Sense Multiple Access with Collision Avoidance (async)
- Access based on sensing bus inactivity during a synchronization interval plus a uniformly distributed random access interval with probability p (p -persistent)
- Not collision-free
- Example: IEEE 802.11



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MAC – CSMA/CA with RTS/CTS

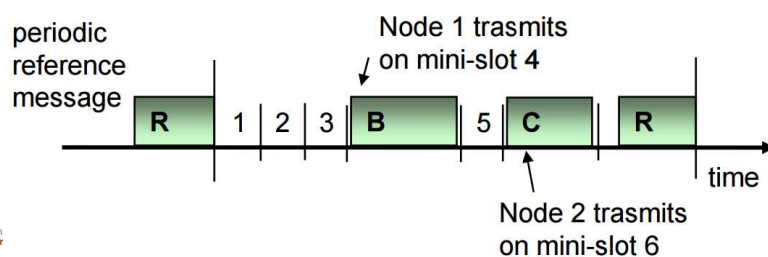
- CSMA/CA: Carrier-Sense Multiple Access with Collision Avoidance using RTS/CTS
- Access requested by sending small packet (RTS).
- Access granted by receiving small packet (CTS).
- Not collision-free on RTS
- RTS sending can use CSMA/CA async.



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MAC – CSMA/CA with sync

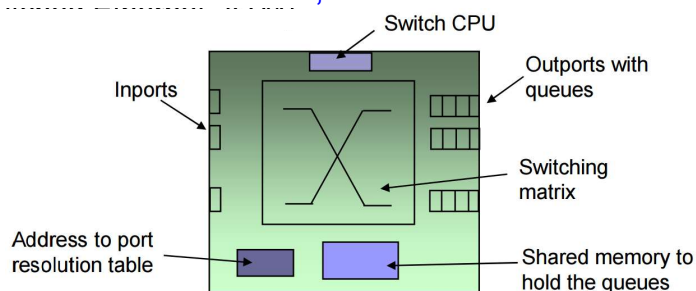
- CSMA/CA Carrier-Sense Multiple Access with Collision Avoidance (with sync)
- Access based on sensing bus inactivity during a number of predefined time-slots (mini-slots) after reception of a synchronous reference message
- Corresponding mini-slot determines priority
- Collision-free and deterministic
- Examples: FlexRay-async (Byteflight), ARINC629-async



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MAC - Switched

- Switched micro-segmented network with central switching hub
- Nodes send asynchronously messages to the switch using point-to-point links
 - no collisions
- Contention at the network access is replaced by contention at the output ports
- Examples: Switched Ethernet, ATM



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MAC - Switched

- The switch handles incoming messages, interprets the destination address and, if known, routes the message to the respective outputport (forwarding)
- For multicasts, broadcasts or unknown addresses, the switch forwards to all outputports (flooding)
- No collisions. Concurrent messages for the same port are queued in memory and sent in sequence
- In principle, it is deterministic but
 - Possible priority inversions in queues!
 - There may be queue overflows!

Medium Access Control

